Antenna Theory and Design

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Abstract

This report presents a structured study portfolio for *Antenna Theory and Design*, developed as part of graduate-level work in Electrical and Computer Engineering at Kennesaw State University. The material is organized chapter by chapter and focuses on the fundamental electromagnetic principles required to understand antenna behavior, including radiation mechanisms, Maxwell's equations, vector potential formulation, ideal dipoles, radiation patterns, directivity, gain, input impedance, radiation efficiency, and polarization.

The main objective of this work is not only to summarize antenna theory but also to connect the physical antenna layer to modern engineering applications, including radar systems, MIMO radar, beamforming, direction-of-arrival estimation, digital signal processing, and FPGA-based signal-processing architectures. Each chapter includes conceptual explanations, mathematical derivations, numerical examples, presentation materials, and solutions to educational problems. Emphasis is placed on understanding how antenna current distributions produce radiated fields and how antenna parameters affect received signal amplitude, phase, array manifold modeling, covariance-matrix formation, and ultimately radar and DOA estimation performance.

The report is based on independent study of Stutzman and Thiele's *Antenna Theory and Design*, 3rd Edition, together with original derivations, examples, figures, and explanatory notes prepared for educational and professional portfolio development. The long-term goal is to build a rigorous bridge between electromagnetic antenna theory and practical RF, radar, MIMO, DSP, and hardware-accelerated signal-processing systems.

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Summary of Chapter 1

1.1 Summary of Chapter 1

The main source for this summary is Chapter 1 of *Antenna Theory and Design*, 3rd edition, by Stutzman and Thiele.

1.1.1 Purpose of Chapter 1

Chapter 1 mainly provides a conceptual roadmap for entering the field of antenna engineering. It does not yet enter heavy mathematical derivations, but it clarifies three fundamental questions:

1. What is an antenna?
2. How does an antenna radiate?
3. What are the main antenna types and the important design parameters?

For my learning path, this chapter should be understood as a bridge between electromagnetics, RF engineering, communication/radar systems, and array signal processing.

1.1.2 What Is an Antenna?

The engineering definition used in this chapter is:

> An antenna is the part of a transmitting or receiving system that is designed to radiate or receive electromagnetic waves.

From the RF system point of view,

$$\text{Guided Wave} \longleftrightarrow \text{Free-Space Wave}$$

This means that an antenna acts as a transducer:

- In the transmitting mode, it converts a guided wave on a cable or transmission line into a free-space electromagnetic wave.
- In the receiving mode, it converts a free-space electromagnetic wave into a guided signal at the input of the receiver.

Therefore, an antenna is not just a piece of metal. It is an interface between a circuit and free space.

1.1.3 Difference Between an Antenna and a Transmission Line

A transmission line, such as a coaxial cable, waveguide, or microstrip line, attempts to keep the electromagnetic energy confined.

An antenna does exactly the opposite. It attempts to detach the energy from the structure and launch it into space.

For a transmission line, the goal is

$$\text{No radiation}$$

For an antenna, the goal is

$$\text{Controlled radiation}$$

From this viewpoint, an antenna can be considered a controlled discontinuity in a wave-guiding system. This discontinuity is intentionally designed so that the electromagnetic fields can separate from the structure and propagate into space.

1.1.4 The Role of Wavelength in Antenna Engineering

The most important scale in antenna engineering is the wavelength:

$$\lambda = \frac{c}{f}$$

In practical form,

$$\lambda(m) = \frac{300}{f(MHz)}$$

and

$$\lambda(cm) = \frac{30}{f(GHz)}$$

An antenna radiates effectively when its physical size is a significant fraction of the wavelength.

For example:

- At low frequencies, the wavelength is large, so it becomes difficult to build an efficient antenna.
- At GHz frequencies, the wavelength becomes smaller, making patch antennas, horn antennas, arrays, and aperture antennas more practical.
- In radar and phased-array systems, the element spacing is often selected around $\lambda/2$.

This point becomes very important later in array antennas, grating lobes, beam steering, and direction-of-arrival estimation.

1.1.5 When Is an Antenna Used?

The book explains that the choice between a cable/transmission line and an antenna is not only about loss. It also depends on the application.

An antenna is necessary when:

- The system is mobile, such as a vehicle, aircraft, satellite, ship, or mobile phone.
- Broadcasting is required, such as radio, television, or cellular base stations.
- Radar or remote sensing is involved.
- Point-to-point communication over a long distance is more economical than using a cable.
- Cable installation is not practical or economical.

In radar systems, the antenna is not only a transmitting device. It determines the direction in which energy is transmitted and the direction from which energy is received. Therefore, in radar, the antenna pattern directly affects detection, resolution, and direction-of-arrival estimation.

1.1.6 How Does an Antenna Radiate?

Chapter 1 explains radiation using an important physical idea:

> Accelerated charges radiate.

In an antenna, charges on a conductor oscillate back and forth because of the RF source. This oscillation means that their velocity and direction are changing, so acceleration exists. The result is a disturbance in the electromagnetic field that separates from the antenna and propagates through space.

A very important point is:

Radiation does not mean that electrons leave the antenna and travel into space. Instead, the electric and magnetic fields detach from the structure and propagate as an electromagnetic wave.

At large distances from the antenna,

$$E, H \propto \frac{1}{r}$$

but the power density behaves as

$$S \propto \frac{1}{r^2}$$

This $1/r^2$ reduction occurs because the radiated power spreads over the surface of a larger sphere, not because the power is destroyed.

1.1.7 Transmission-Line View of a Dipole

One of the best explanations in this chapter is the following:

If the open end of a transmission line is bent outward, a structure similar to a dipole antenna is obtained.

In an ordinary transmission line, the currents on the two conductors flow in opposite directions, and the external fields largely cancel each other.

However, when the two branches are opened outward, the currents are arranged in a way that allows the fields to couple to free space.

This viewpoint is also important for understanding patch antennas. A patch antenna can also be seen as a resonant structure in which the fields at the edges create fringing fields, and these fringing fields are responsible for radiation.

1.1.8 Main Antenna Parameters

Chapter 1 introduces several key parameters. These parameters form the backbone of the entire course.

Radiation Pattern

The radiation pattern describes the angular variation of radiation from an antenna:

$$F(\theta, \phi)$$

It shows the directions in which the antenna radiates more strongly or more weakly.

In radar and direction-of-arrival estimation, the radiation pattern determines how the array samples space.

Directivity

Directivity shows how strongly an antenna concentrates radiated power in a particular direction.

If a hypothetical isotropic antenna distributes power equally in all directions, a directive antenna concentrates the same power in a preferred direction.

Gain

Gain is similar to directivity, but it also includes the real losses of the antenna:

$$G = e_r D$$

where

- D is the directivity,
- e_r is the radiation efficiency,
- G is the gain.

Therefore, an antenna may have good directivity, but if its loss is high, its actual gain will be low.

Polarization

Polarization describes the direction and time behavior of the electric-field vector of the radiated wave.

The main types are:

- Linear polarization,
- Circular polarization,
- Elliptical polarization.

In practical systems, polarization mismatch can reduce received power.

Input Impedance

The input impedance of an antenna is

$$Z_A = R_A + jX_A$$

For a good connection to a transmission line, the antenna impedance should be matched to the system impedance, usually 50 Ω.

This topic will later connect directly to S_{11}, return loss, VSWR, and the Smith chart.

Bandwidth

Bandwidth is the frequency range over which the antenna has acceptable performance.

Acceptable performance can be defined in terms of:

- S_{11},
- gain,
- radiation pattern,
- polarization,
- efficiency.

Therefore, bandwidth is not just a simple number. It depends on the design criterion.

Scanning

Scanning means moving the antenna beam in space.

This can be done:

- mechanically, such as rotating a dish antenna,
- electronically, such as in a phased array.

For my research path, this part is very important because it connects directly to beamforming and direction-of-arrival estimation.

1.1.9 The Four Main Types of Antennas

Chapter 1 divides antennas into four main families.

Electrically Small Antennas

These antennas are much smaller than the wavelength:

$$L \ll \lambda$$

Their main characteristics are:

- low directivity,
- low radiation resistance,
- high reactance,
- usually low efficiency.

Examples include:

- short dipole,
- small loop,
- short AM receiving antenna on a car.

The main problem with these antennas is that, because they are small, they do not radiate efficiently and are difficult to match.

Resonant Antennas

These antennas operate well at a single frequency or within a narrow frequency band.

Examples include:

- half-wave dipole,

- monopole,
- microstrip patch,
- Yagi-Uda antenna.

Their main characteristics are:

- simpler structure,
- low to moderate gain,
- limited bandwidth,
- impedance that is closer to a real value and easier to match.

For this course, the rectangular microstrip patch antenna is a very important example of a resonant antenna.

Broadband Antennas

These antennas have acceptable performance over a wide frequency range.

Examples include:

- spiral antenna,
- log-periodic antenna,
- biconical antenna.

Their main characteristics are:

- wide bandwidth,
- usually low to moderate gain,
- relatively stable impedance,
- useful for wideband applications and antenna measurements.

An important idea in the chapter is that a broadband antenna usually has an active region. This means that, at each frequency, only a certain portion of the antenna radiates effectively.

Aperture Antennas

These antennas have a physical opening through which the wave passes or is focused.

Examples include:

- horn antenna,
- parabolic reflector,
- lens antenna.

Their main characteristics are:

- high gain,
- narrow beam,
- suitable for microwave, radar, and satellite communication systems,
- the larger the aperture is relative to the wavelength, the narrower the beam and the higher the gain.

In radar systems, aperture antennas are very important because they provide high gain and narrow beams.

1.1.10 Choosing an Antenna Based on Gain

Chapter 1 provides a useful engineering guide for approximate antenna gain.

Approximate Gain	Suitable Antenna Type
Less than 5 dB	Electrically small antennas, loops, dipoles
5 to 8 dB	Microstrip patches, spirals
8 to 15 dB	Yagi-Uda, helix, log-periodic antennas
More than 15 dB	Horn, reflector, and aperture antennas

This table is very useful for practical design. For example, if a project requires a gain higher than 20 dB, a single patch antenna is usually not enough. In that case, an array or an aperture antenna is typically required.

1.1.11 Main Message of the Chapter

If I summarize Chapter 1 in a few engineering sentences:

An antenna is not an isolated component. It is a physical circuit-spatial energy conversion system. Its performance is defined by wavelength, surface current, near-field, far-field, impedance, radiation pattern, gain, polarization, and bandwidth. No antenna is best in all parameters. Antenna design is always a trade-off among size, bandwidth, gain, efficiency, impedance matching, cost, and system application.

1.1.12 Important Week 1 Points to Remember

For quick review, the following points should be remembered:

1. An antenna converts a guided wave into a radiated wave.
2. Radiation is produced by accelerated charges and oscillating currents.
3. Wavelength is the main design scale in antenna engineering.
4. In the far field, the fields behave locally like a plane wave.
5. The radiation pattern is the most important spatial description of an antenna.
6. Gain is equal to directivity multiplied by efficiency.

$$G = e_r D$$

7. Impedance matching determines how much power enters the antenna.
8. Bandwidth means acceptable performance, not only acceptable S_{11}.
9. The four main antenna families are electrically small antennas, resonant antennas, broadband antennas, and aperture antennas.
10. For radar and direction-of-arrival estimation, the radiation pattern and array behavior are as important as the DSP algorithm.

Very Short Final Summary

Chapter 1 aims to move the student's thinking from "an antenna as a piece of metal" to "an antenna as a physical-circuit-spatial system." For my research direction, the most important lesson is that the performance of MIMO radar and direction-of-arrival algorithms does not depend only on MUSIC, covariance matrices, or FPGA implementation. It is shaped and limited from the beginning by antenna pattern, element spacing, polarization, gain, impedance, and mutual coupling.

1.2 Eamples

Numerical Example of Basic Antenna Parameters

Problem Statement Consider a thin, center-fed, vertical half-wave dipole antenna operating in free space at

$$f_0 = 2.4 \text{ GHz}.$$

Assume that the antenna is used in a 50 Ω RF system. The antenna is approximately resonant at the operating frequency, and its radiation efficiency is assumed to be

$$e_r = 0.90.$$

For this antenna, determine and explain the following basic antenna parameters:

1. Wavelength and approximate physical length,
2. Radiation pattern,
3. Half-power beamwidth,
4. Directivity,
5. Gain,
6. Polarization,
7. Input impedance,
8. Reflection coefficient, return loss, VSWR, and mismatch loss,
9. Bandwidth,
10. Engineering interpretation of the results.

Solution

We use a half-wave dipole antenna as a simple numerical example because it clearly illustrates the most important antenna parameters: radiation pattern, directivity, gain, polarization, input impedance, and bandwidth.

1. Frequency and Wavelength

The operating frequency is

$$f_0 = 2.4 \text{ GHz}.$$

The speed of light in free space is approximately

$$c = 3 \times 10^8 \text{ m/s}.$$

The wavelength is

$$\lambda = \frac{c}{f_0}.$$

Substituting the numerical values,

$$\lambda = \frac{3 \times 10^8}{2.4 \times 10^9} = 0.125 \text{ m}.$$

Therefore,

$$\lambda = 12.5 \text{ cm}.$$

For an ideal half-wave dipole, the physical length is approximately

$$L \approx \frac{\lambda}{2}.$$

Thus,

$$L \approx \frac{12.5 \text{ cm}}{2} = 6.25 \text{ cm}.$$

In practice, a real half-wave dipole is usually slightly shorter than $\lambda/2$, often around

$$L \approx 0.48\lambda.$$

Therefore,

$$L \approx 0.48(12.5 \text{ cm}) = 6.0 \text{ cm}.$$

So the practical antenna length is approximately

$$L \approx 6.0 \text{ cm}.$$

2. Radiation Pattern

Assume that the dipole is oriented vertically along the z-axis.

For a vertical half-wave dipole:

- Maximum radiation occurs in the direction perpendicular to the antenna axis.
- Minimum radiation, or nulls, occur along the antenna axis.

Therefore, the maximum radiation occurs in the horizontal plane:

$$\theta = 90^\circ.$$

The nulls occur at

$$\theta = 0^\circ$$

and

$$\theta = 180^\circ.$$

The approximate normalized field pattern of a half-wave dipole is

$$F(\theta) = \frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta}.$$

This is the normalized field pattern as a function of the elevation angle θ.

At the direction of maximum radiation,

$$F(90^\circ) = 1.$$

This means that the radiation is maximum in the plane perpendicular to the dipole.

The pattern of a vertical half-wave dipole is often described as a doughnut-shaped pattern. It is strong around the sides of the antenna and weak along the top and bottom directions.

3. Half-Power Beamwidth

The half-power points are the angular positions where the radiated power drops to one-half of its maximum value.

Since power is proportional to the square of the field magnitude,

$$P(\theta) \propto |F(\theta)|^2.$$

The half-power condition is

$$|F(\theta)|^2 = \frac{1}{2}.$$

Equivalently,

$$|F(\theta)| = \frac{1}{\sqrt{2}} \approx 0.707.$$

For a half-wave dipole, the half-power points occur approximately at

$$\theta \approx 51^\circ$$

and

$$\theta \approx 129^\circ.$$

Therefore, the half-power beamwidth is

$$HPBW = 129^\circ - 51^\circ.$$

Thus,

$$HPBW \approx 78^\circ.$$

This indicates that a half-wave dipole is not a highly directive antenna. Its beam is relatively broad.

4. Directivity

The directivity of a half-wave dipole is approximately

$$D \approx 1.64.$$

In decibels relative to an isotropic radiator,

$$D_{\text{dBi}} = 10 \log_{10}(D).$$

Substituting,

$$D_{\text{dBi}} = 10 \log_{10}(1.64).$$

Therefore,

$$D_{\mathrm{dBi}} \approx 2.15 \text{ dBi}.$$

So the directivity of the half-wave dipole is

$$D = 1.64$$

or

$$D = 2.15 \text{ dBi}.$$

This means that, in its maximum radiation direction, the half-wave dipole concentrates power approximately 1.64 times more strongly than an ideal isotropic antenna radiating the same total power.

Directivity depends only on the shape of the radiation pattern. It does not include antenna losses.

5. Gain

The gain of an antenna includes both directivity and radiation efficiency.

The relationship between gain, directivity, and radiation efficiency is

$$G = e_r D.$$

Given

$$e_r = 0.90$$

and

$$D = 1.64,$$

we obtain

$$G = 0.90 \times 1.64.$$

Therefore,

$$G = 1.476.$$

In dB,

$$G_{\text{dBi}} = 10 \log_{10}(1.476).$$

Thus,

$$G_{\text{dBi}} \approx 1.69 \text{ dBi}.$$

So the gain is approximately

$$G \approx 1.48$$

or

$$G \approx 1.69 \text{ dBi}.$$

If the antenna accepts

$$P_{\text{in}} = 1 \text{ W}$$

at its input terminals, the radiated power is

$$P_{\text{rad}} = e_r P_{\text{in}}.$$

Therefore,

$$P_{\text{rad}} = 0.90 \times 1.$$

Thus,

$$P_{\text{rad}} = 0.90 \text{ W}.$$

This means that if 1 W is accepted by the antenna, approximately 0.90 W is actually radiated into space, while 0.10 W is lost in the antenna structure.

6. Polarization

If the dipole is installed vertically, the radiated electric field is also vertically polarized.

Therefore, the antenna polarization is

$$\text{Linear Vertical Polarization.}$$

If the receiving antenna is also vertically polarized, the polarization match is good.

If the receiving antenna is horizontally polarized, then in the ideal case the received power becomes approximately zero because of polarization mismatch.

If the angle between the transmitting and receiving linear polarizations is ψ, the polarization loss factor is approximately

$$PLF = \cos^2(\psi).$$

For example, if

$$\psi = 45^\circ,$$

then

$$PLF = \cos^2(45^\circ).$$

Since

$$\cos(45^\circ) = \frac{1}{\sqrt{2}},$$

we get

$$PLF = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}.$$

Therefore,

$$PLF = 0.5.$$

In decibels,

$$PLF_{\text{dB}} = 10\log_{10}(0.5).$$

Thus,

$$PLF_{\text{dB}} \approx -3 \text{ dB}.$$

So a 45° polarization mismatch causes a 3 dB power loss.

This is very important in radar and communication systems. Even if the antenna has good gain, polarization mismatch can significantly reduce the received power.

7. Input Impedance

For a thin, center-fed, resonant half-wave dipole in free space, the input impedance is approximately

$$Z_A \approx 73 + j0 \ \Omega.$$

Therefore,

$$R_A \approx 73 \ \Omega$$

and

$$X_A \approx 0 \ \Omega.$$

The real part represents the input resistance, and the imaginary part represents the input reactance.

At resonance, the reactance is approximately zero:

$$X_A \approx 0.$$

Now assume that the RF system impedance is

$$Z_0 = 50\ \Omega.$$

The antenna is not perfectly matched to the 50 Ω system, because

$$Z_A \neq Z_0.$$

8. Reflection Coefficient

The voltage reflection coefficient is

$$\Gamma = \frac{Z_A - Z_0}{Z_A + Z_0}.$$

Using

$$Z_A = 73\ \Omega$$

and

$$Z_0 = 50\ \Omega,$$

we get

$$\Gamma = \frac{73 - 50}{73 + 50}.$$

Thus,

$$\Gamma = \frac{23}{123}.$$

Therefore,

$$\Gamma \approx 0.187.$$

The magnitude of the reflection coefficient is

$$|\Gamma| \approx 0.187.$$

9. Return Loss

Return loss is defined as

$$RL = -20 \log_{10} |\Gamma|.$$

Substituting

$$|\Gamma| = 0.187,$$

we get

$$RL = -20 \log_{10}(0.187).$$

Therefore,

$$RL \approx 14.6 \text{ dB}.$$

This is a reasonably good match for many practical systems.

10. VSWR

The voltage standing wave ratio is

$$VSWR = \frac{1+|\Gamma|}{1-|\Gamma|}.$$

Substituting

$$|\Gamma| = 0.187,$$

we obtain

$$VSWR = \frac{1+0.187}{1-0.187}.$$

Thus,

$$VSWR = \frac{1.187}{0.813}.$$

Therefore,

$$VSWR \approx 1.46.$$

A VSWR of approximately 1.46 is acceptable in many RF systems, although it is not a perfect match.

11. Accepted Power and Reflected Power

The accepted power is related to the incident power by

$$P_{\text{accepted}} = P_{\text{incident}}(1-|\Gamma|^2).$$

Assume

$$P_{\text{incident}} = 1 \text{ W}.$$

Then

$$P_{\text{accepted}} = 1(1 - 0.187^2).$$

Since

$$0.187^2 \approx 0.035,$$

we get

$$P_{\text{accepted}} \approx 1(1 - 0.035).$$

Therefore,

$$P_{\text{accepted}} \approx 0.965 \text{ W}.$$

So approximately 96.5% of the incident power is accepted by the antenna, and approximately 3.5% is reflected back toward the source.

The reflected power is

$$P_{\text{reflected}} = P_{\text{incident}}|\Gamma|^2.$$

Thus,

$$P_{\text{reflected}} = 1(0.187^2) \approx 0.035 \text{ W}.$$

Therefore,

$$P_{\text{reflected}} \approx 35 \text{ mW}.$$

12. Mismatch Loss

Mismatch loss is defined as

$$ML = -10\log_{10}(1 - |\Gamma|^2).$$

Substituting

$$|\Gamma| = 0.187,$$

we get

$$ML = -10\log_{10}(1 - 0.187^2).$$

Therefore,

$$ML \approx 0.15 \text{ dB}.$$

This means that the mismatch between the 73 Ω dipole and the 50 Ω system causes only about 0.15 dB of mismatch loss.

13. Bandwidth

Bandwidth must always be defined using a specific performance criterion.

A common impedance-bandwidth criterion is

$$S_{11} < -10 \text{ dB}.$$

This means that the reflection coefficient is sufficiently small and that the antenna is reasonably well matched over that frequency range.

Assume that the measured or simulated frequency range satisfying

$$S_{11} < -10 \text{ dB}$$

is

$$f_L = 2.34 \text{ GHz}$$

to

$$f_H = 2.46 \text{ GHz}.$$

The absolute bandwidth is

$$BW = f_H - f_L.$$

Substituting,

$$BW = 2.46 - 2.34.$$

Therefore,

$$BW = 0.12 \text{ GHz}.$$

Thus,

$$BW = 120 \text{ MHz}.$$

The fractional bandwidth is

$$FBW = \frac{BW}{f_0}.$$

Using

$$BW = 0.12 \text{ GHz}$$

and

$$f_0 = 2.4 \text{ GHz},$$

we get

$$FBW = \frac{0.12}{2.4}.$$

Therefore,

$$FBW = 0.05.$$

In percentage form,

$$FBW = 5\%.$$

So the antenna has approximately

$$5\%$$

fractional bandwidth based on the $S_{11} < -10$ dB criterion.

14. Numerical Summary

Parameter	**Numerical Value**	**Interpretation**
Operating Frequency	2.4 GHz	Center frequency
Wavelength	12.5 cm	Main physical design scale
Practical Dipole Length	≈ 6.0 cm	Approximately 0.48λ
Radiation Pattern	Doughnut-shaped	Maximum radiation broadside to the antenna
Nulls	$\theta = 0°, 180°$	Almost no radiation along the antenna axis
HPBW	$\approx 78°$	Half-power beamwidth
Directivity	$1.64 = 2.15$ dBi	Spatial concentration of radiated power
Efficiency	90%	Assumed radiation efficiency
Gain	$1.48 = 1.69$ dBi	Directivity including radiation efficiency
Polarization	Linear vertical	For a vertically mounted dipole
Input Impedance	$73 + j0\ \Omega$	Approximate resonant impedance
$\lvert\Gamma\rvert$	0.187	Voltage reflection coefficient magnitude
Return Loss	14.6 dB	Reasonable impedance match
VSWR	1.46	Acceptable standing-wave ratio
Mismatch Loss	0.15 dB	Small mismatch loss
Bandwidth	120 MHz	For $S_{11} < -10$ dB
Fractional Bandwidth	5%	Relative bandwidth

15. Engineering Interpretation

This example shows how the main antenna parameters are connected.

First, the radiation pattern tells us where the antenna radiates strongly and where it radiates weakly. For a vertical half-wave dipole, radiation is strongest in the horizontal plane and weakest along the antenna axis.

Second, directivity describes only the spatial concentration of radiation. For a half-wave dipole, the directivity is approximately

$$D = 1.64 = 2.15 \text{ dBi}.$$

This means that the dipole is more directive than an isotropic radiator, but it is still a relatively broad-beam antenna.

Third, gain includes the effect of radiation efficiency. Since the radiation efficiency is assumed to be

$$e_r = 0.90,$$

the gain is lower than the directivity:

$$G = e_r D = 1.476 = 1.69 \text{ dBi}.$$

Therefore, loss in the antenna reduces the useful radiated performance.

Fourth, polarization is important because the transmitting and receiving antennas must have compatible electric-field orientations. A 45° polarization mismatch causes a 3 dB power loss, and a 90° mismatch ideally causes complete loss of received signal.

Fifth, input impedance determines how much power can be delivered from the RF system to the antenna. A resonant half-wave dipole has approximately

$$Z_A = 73 + j0 \ \Omega.$$

When connected to a 50 Ω system, the antenna is not perfectly matched, but the mismatch is moderate:

$$|\Gamma| \approx 0.187,$$

$$RL \approx 14.6 \text{ dB},$$

and

$$VSWR \approx 1.46.$$

This means that most of the incident power is accepted by the antenna.

Sixth, bandwidth must be defined with respect to a specific criterion. In this example, the impedance bandwidth is defined by

$$S_{11} < -10 \text{ dB}.$$

Using the assumed frequency range from 2.34 GHz to 2.46 GHz, the bandwidth is

$$BW = 120 \text{ MHz}$$

and the fractional bandwidth is

$$FBW = 5\%.$$

16. Connection to Radar, Arrays, and Direction-of-Arrival Estimation

In radar arrays and direction-of-arrival estimation, each antenna element is not just an ideal point sensor. Each element has its own radiation pattern, gain, polarization, impedance, bandwidth, and efficiency.

A useful conceptual expression is

$$\text{Measured Array Response} = \text{Element Pattern} \times \text{Array Factor}.$$

Therefore, if a MUSIC, ESPRIT, or beamforming algorithm is simulated using ideal isotropic elements, the result may look very clean. However, in real hardware, the antenna parameters affect the received signal.

In practical systems, antenna parameters can cause:

- SNR reduction,

- angular bias,
- gain variation with angle,
- polarization loss,
- impedance mismatch,
- mutual coupling,
- bandwidth limitations,
- distortion of the array manifold.

Therefore, understanding antenna parameters is essential not only for antenna design, but also for radar signal processing and direction-of-arrival estimation.

Final Summary

A half-wave dipole at 2.4 GHz has a wavelength of 12.5 cm, a practical length of about 6.0 cm, a doughnut-shaped radiation pattern, a directivity of approximately 2.15 dBi, and a gain of approximately 1.69 dBi when the radiation efficiency is 90%. Its resonant input impedance is approximately $73 + j0\ \Omega$, which gives a return loss of about 14.6 dB and a VSWR of about 1.46 when connected to a 50 Ω system. If its $S_{11} < -10$ dB frequency range is from 2.34 GHz to 2.46 GHz, its impedance bandwidth is 120 MHz, or 5% fractional bandwidth.

This example demonstrates that antenna performance is not described by a single number. A complete antenna description requires radiation pattern, directivity, gain, polarization, impedance matching, efficiency, and bandwidth together.

1.3 Problems

Problem Show that the decay of power density for a wave radiated by an antenna, proportional to $1/r^2$, results from the increase in the area of a sphere centered on the antenna and from the fact that the total radiated power is conserved.

Solution Assume that an antenna radiates a total power

$$P_{\text{rad}}.$$

At a distance r from the antenna, imagine a sphere centered on the antenna. The surface area of this sphere is

$$A = 4\pi r^2.$$

If the antenna were an ideal isotropic radiator, it would radiate power equally in all directions. Therefore, the radiated power would be uniformly distributed over the surface of the sphere.

The average power density on the sphere is

$$S_{\text{avg}}(r) = \frac{P_{\text{rad}}}{4\pi r^2}.$$

Thus,

$$S_{\text{avg}}(r) \propto \frac{1}{r^2}.$$

This shows that the power density decreases as $1/r^2$ because the same total radiated power spreads over a spherical area that increases as r^2.

General Directional Case

For a real antenna, radiation is not necessarily uniform in all directions. In this case, the power density may depend on the angular direction (θ, ϕ). We can write

$$S(r, \theta, \phi) = \frac{U(\theta, \phi)}{r^2},$$

where $U(\theta, \phi)$ is the radiation intensity, measured in watts per steradian.

The total radiated power is obtained by integrating the power density over a spherical surface:

$$P_{\text{rad}} = \int_0^{2\pi} \int_0^{\pi} S(r, \theta, \phi) r^2 \sin\theta \, d\theta \, d\phi.$$

Substituting

$$S(r, \theta, \phi) = \frac{U(\theta, \phi)}{r^2},$$

we obtain

$$P_{\text{rad}} = \int_0^{2\pi} \int_0^{\pi} \frac{U(\theta, \phi)}{r^2} r^2 \sin\theta \, d\theta \, d\phi.$$

The factors r^2 cancel:

$$P_{\text{rad}} = \int_0^{2\pi} \int_0^{\pi} U(\theta, \phi) \sin\theta \, d\theta \, d\phi.$$

Therefore, the total radiated power is independent of r. This is consistent with conservation of power.

Physical Interpretation

The antenna radiates electromagnetic power into space. As the wave propagates outward, the total power crossing any spherical surface centered on the antenna remains the same, assuming a lossless medium.

However, the area of the spherical surface increases as $4\pi r^2$. Therefore, the power per unit area must decrease as $\frac{1}{r^2}$. This is the physical origin of the $1/r^2$ power-density decay.

Connection to Field Strength

Since power density is proportional to the square of the electric-field magnitude,

$$S \propto |E|^2,$$

and

$$S \propto \frac{1}{r^2},$$

it follows that

$$|E| \propto \frac{1}{r}.$$

Similarly,

$$|H| \propto \frac{1}{r}.$$

Thus, in the far field of an antenna,

$$E, H \propto \frac{1}{r},$$

while

$$S \propto \frac{1}{r^2}.$$

Final Answer The $1/r^2$ decay of radiated power density is a direct result of conservation of total radiated power and the spherical spreading of the wave. Since the surface area of a sphere increases as $4\pi r^2$, the power per unit area must decrease as $1/r^2$.

$$\boxed{S(r) = \frac{P_{\text{rad}}}{4\pi r^2}}$$

Therefore,

$$\boxed{S(r) \propto \frac{1}{r^2}.}$$

1.3.1 Problem

Problem Statement: Consider an ideal dipole of length $L \ll \lambda$ If charge carriers actually moved along the full length of the wire in half a cycle, $T/2$, show that the speed of the charges is much less than the speed of light; that is, $v \ll c$

Solution: The period of the sinusoidal excitation is

$$T = \frac{1}{f},$$

where f is the operating frequency.

The wavelength is related to frequency by

$$\lambda = \frac{c}{f},$$

or equivalently,

$$f = \frac{c}{\lambda}.$$

According to the problem statement, suppose that the charge carriers move along the full length L of the wire during half of one cycle:

$$\Delta t = \frac{T}{2}.$$

The average speed of the charge carriers would then be

$$v = \frac{\text{distance traveled}}{\text{time interval}}.$$

Thus,

$$v = \frac{L}{T/2}.$$

Therefore,

$$v = \frac{2L}{T}.$$

Since

$$T = \frac{1}{f},$$

we have

$$v = 2Lf.$$

Using

$$f = \frac{c}{\lambda},$$

we obtain

$$v = 2L\frac{c}{\lambda}.$$

Thus,

$$v = c\left(\frac{2L}{\lambda}\right).$$

Dividing both sides by c,

$$\frac{v}{c} = \frac{2L}{\lambda}.$$

Since the dipole is electrically very short,

$$L \ll \lambda.$$

Therefore,

$$\frac{L}{\lambda} \ll 1.$$

Consequently,

$$\frac{2L}{\lambda} \ll 1.$$

Thus,

$$\frac{v}{c} \ll 1.$$

Therefore,

$$v \ll c.$$

Final Answer

The average charge-carrier speed would be

$$\boxed{v = \frac{2L}{T}}$$

or

$$\boxed{v = 2Lf.}$$

Using

$$f = \frac{c}{\lambda},$$

we get

$$\boxed{v = c\left(\frac{2L}{\lambda}\right).}$$

Since

$$L \ll \lambda,$$

then

$$\boxed{\frac{v}{c} = \frac{2L}{\lambda} \ll 1.}$$

Therefore,

$$\boxed{v \ll c.}$$

Physical Interpretation

This result shows that the actual charge carriers do not need to travel at the speed of light for electromagnetic radiation to occur. The electromagnetic disturbance propagates at approximately the speed of light, but the charges themselves only undergo small oscillatory motion.

This is similar to a wave traveling along a rope: the wave can propagate rapidly, while the individual particles of the rope move only locally.

In antennas, it is therefore more accurate to think of the current as an oscillating charge disturbance rather than as electrons physically traveling large distances at relativistic speeds.

1.3.2 Problem

Problem Statement: Project: Antenna hunt.

1. Locate one representative antenna for each of the four major antenna-type categories. Preferably, the antennas will be in your community and be actively used. Take a photograph, or make a sketch, of the antenna and its surroundings. If you cannot find an example in your community, you may use other sources such as a catalog, magazine, or the Internet.

2. Prepare a brief report that shows each of the four antennas and includes information about the antenna such as the type of antenna, operating frequency, purpose and use, and any other information you can determine.

Solution: The four major antenna categories introduced in Chapter 1 are:

1. Electrically small antennas,
2. Resonant antennas,
3. Broadband antennas,
4. Aperture antennas.

A sample antenna-hunt report is presented below.

Sample Antenna-Hunt Report

1. Electrically Small Antenna

Example: Car AM Radio Monopole Antenna A common example of an electrically small antenna is the vertical monopole antenna used for AM radio reception in automobiles.

Antenna type: Electrically small monopole

The AM broadcast band is approximately

$$f = 0.53 \text{ MHz to } 1.7 \text{ MHz}.$$

A representative frequency is

$$f = 1.0 \text{ MHz}.$$

The corresponding wavelength is

$$\lambda = \frac{c}{f}.$$

Using

$$c = 3 \times 10^8 \text{ m/s}$$

and

$$f = 1.0 \times 10^6 \text{ Hz},$$

we get

$$\lambda = \frac{3 \times 10^8}{1.0 \times 10^6} = 300 \text{ m}.$$

A typical car antenna length is about

$$L \approx 1 \text{ m}.$$

Thus,

$$\frac{L}{\lambda} = \frac{1}{300} \approx 0.0033.$$

Since

$$L \ll \lambda,$$

this antenna is electrically small.

Purpose and Use This antenna is used to receive AM broadcast radio signals in vehicles.

Main Characteristics

- Very small compared with wavelength,
- Nearly omnidirectional in the horizontal plane,

- Low radiation resistance,
- High reactance,
- Low efficiency when used for transmission,
- Acceptable for reception because received-signal power requirements are small.

Sketch Placeholder

Insert photo or sketch of a car AM monopole antenna here.

2. Resonant Antenna

Example: Half-Wave Dipole Antenna A half-wave dipole is one of the most important examples of a resonant antenna.

Antenna type: Resonant antenna

Assume a half-wave dipole operating at

$$f = 100 \text{ MHz}.$$

The wavelength is

$$\lambda = \frac{c}{f}.$$

Substituting,

$$\lambda = \frac{3 \times 10^8}{100 \times 10^6} = 3 \text{ m}.$$

The approximate half-wave dipole length is

$$L \approx \frac{\lambda}{2}.$$

Therefore,

$$L \approx \frac{3}{2} = 1.5 \text{ m}.$$

In practice, the length is often slightly shorter than $\lambda/2$, due to end effects.

Purpose and Use Half-wave dipoles are used in communication systems, laboratory measurements, amateur radio, and as reference antennas.

Main Characteristics

- Operates efficiently near its resonant frequency,
- Has a real-valued input impedance near resonance,
- Typical input impedance is approximately $73\ \Omega$ in free space,
- Has a doughnut-shaped radiation pattern,
- Has low to moderate gain,
- Has relatively narrow bandwidth.

Approximate Directivity The directivity of a half-wave dipole is approximately

$$D \approx 1.64.$$

In decibels,

$$D_{\text{dBi}} = 10 \log_{10}(1.64) \approx 2.15 \text{ dBi}.$$

Sketch Placeholder

Insert photo or sketch of a half-wave dipole antenna here.

3. Broadband Antenna

Example: Log-Periodic Dipole Array A log-periodic dipole array is a common broadband antenna.

Antenna type: Broadband antenna

A typical log-periodic antenna may operate over a wide frequency range, for example,

$$f_L = 50 \text{ MHz}$$

to

$$f_H = 500 \text{ MHz}.$$

The bandwidth ratio is

$$\frac{f_H}{f_L} = \frac{500}{50} = 10.$$

Thus, the antenna has a

$$10 : 1$$

frequency range.

Purpose and Use Log-periodic antennas are used for wideband communication, television reception, electromagnetic compatibility testing, and measurement systems.

Main Characteristics

- Wide operating bandwidth,
- Frequency-independent behavior over a large range,
- Moderate gain,

- Directional radiation pattern,
- Active region shifts along the antenna as frequency changes,
- More stable impedance than a narrowband resonant antenna over a broad band.

Active Region Concept In a broadband antenna, the entire physical structure does not radiate equally at all frequencies. Instead, for each frequency, a particular part of the antenna becomes the active region.

For a log-periodic antenna, lower frequencies generally use the longer elements, while higher frequencies use the shorter elements.

Sketch Placeholder

Insert photo or sketch of a log-periodic dipole array here.

4. Aperture Antenna

Example: Parabolic Reflector Antenna A parabolic dish antenna is a common example of an aperture antenna.

Antenna type: Aperture antenna

Assume a satellite dish antenna operating at

$$f = 12 \text{ GHz}.$$

The wavelength is

$$\lambda = \frac{c}{f}.$$

Substituting,

$$\lambda = \frac{3 \times 10^8}{12 \times 10^9} = 0.025 \text{ m}.$$

Thus,

$$\lambda = 2.5 \text{ cm}.$$

Assume the dish diameter is

$$D = 0.6 \text{ m}.$$

The diameter in wavelengths is

$$\frac{D}{\lambda} = \frac{0.6}{0.025} = 24.$$

Therefore, the aperture is many wavelengths across, which is typical of high-gain aperture antennas.

Purpose and Use Parabolic reflector antennas are used in satellite communication, radar, radio astronomy, and point-to-point microwave links.

Main Characteristics

- High gain,
- Narrow main beam,
- Good directivity,
- Suitable for microwave and millimeter-wave frequencies,
- Gain increases as the aperture becomes larger relative to wavelength.

Approximate Gain Estimate The approximate gain of a parabolic reflector antenna is

$$G \approx \eta_{\text{ap}} \left(\frac{\pi D}{\lambda} \right)^2,$$

where η_{ap} is the aperture efficiency.

Assume

$$\eta_{\text{ap}} = 0.60.$$

Then

$$G \approx 0.60 \left(\frac{\pi(0.6)}{0.025} \right)^2 .$$

First,

$$\frac{\pi(0.6)}{0.025} = 75.4.$$

Therefore,

$$G \approx 0.60(75.4)^2.$$

$$G \approx 0.60(5685).$$

$$G \approx 3411.$$

In dBi,

$$G_{\text{dBi}} = 10 \log_{10}(3411).$$

Thus,

$$G_{\text{dBi}} \approx 35.3 \text{ dBi}.$$

This shows why aperture antennas are widely used when high gain is required.

Sketch Placeholder

Insert photo or sketch of a parabolic reflector antenna here.

Summary Table

Table 1.1: Antenna Categories and Characteristics

Category	Example	Typical Frequency	Main Feature	Common Use
Electrically Small	Car AM Monopole	~ 1 MHz	$L \ll \lambda$	AM radio reception
Resonant	Half-Wave Dipole	~ 100 MHz	$L \approx \lambda/2$	Communication, reference antenna
Broadband	Log-Periodic Array	50–500 MHz	Wide bandwidth	TV, EMC, measurements
Aperture	Parabolic Reflector	~ 12 GHz	$D \gg \lambda$	Satellite, radar, microwave links

Engineering Discussion

The four antennas represent four different design philosophies.

An electrically small antenna is physically much smaller than the wavelength. It is useful when size constraints dominate, but it usually suffers from low radiation resistance and low efficiency.

A resonant antenna uses a physical dimension comparable to a fraction of the wavelength, often around $\lambda/2$. It provides efficient radiation near its resonant frequency but usually has limited bandwidth.

A broadband antenna is designed to maintain acceptable impedance, gain, and radiation pattern over a wide frequency range. Its active radiating region changes with frequency.

An aperture antenna uses a physical opening or reflecting surface that is many wavelengths in size. This allows the antenna to produce high gain and a narrow beam, which is especially important in radar and satellite communication systems.

Final Answer

Representative antennas for the four main antenna categories are:

- $\boxed{\text{Electrically small antenna: Car AM monopole}}$
- $\boxed{\text{Resonant antenna: Half-wave dipole}}$
- $\boxed{\text{Broadband antenna: Log-periodic dipole array}}$
- $\boxed{\text{Aperture antenna: Parabolic reflector antenna}}$

These examples demonstrate that antenna selection depends on operating frequency, wavelength, gain requirement, bandwidth requirement, physical size, and system application.

Overall Chapter 1 Problem Summary

The first problem shows that electromagnetic power density decreases as $1/r^2$ because the same total radiated power spreads over the surface of a sphere whose area increases as $4\pi r^2$.

The second problem shows that charge carriers in an electrically small dipole move much more slowly than the speed of light, even though the electromagnetic disturbance propagates at approximately that speed.

The third problem connects antenna theory to practical antennas in real systems. It shows how electrically small, resonant, broadband, and aperture antennas differ in size, bandwidth, gain, and application.

These three problems together reinforce the main conceptual message of Chapter 1:

$$\boxed{\text{An antenna is a physical interface between circuits and free-space electromagnetic waves.}}$$

Understanding antennas requires connecting wavelength, radiation, power flow, physical size, bandwidth, gain, and system application.